Audit, step 3, area D: app, documentation, integrity, packaging

Auditor: an independent agent (model: Fable), read-only. HEAD 2734546, 2026-10-20 (night). Its findings are below, with the fix status recorded in `DECISIONS.md` (“Audit actions”).

ID	Severity	Status	Location	Finding
D1	High	CONFIRMED	`validation/*` , `tests/run.js`	Committed report and status never regenerated after the V15–V20 aggregation (still V1–V14, git 5f60ace). The aggregation reads `sim/out/loop_air_3T/` , which did not exist yet, so V15 would read NOT RUN rather than FAIL
D2	High	CONFIRMED	`app/index.html`	The Pair explorer and the array solver share one worker. A solve or Cancel during a pair scan kills it silently, and `pairBusy` stays true for the session
D3	Medium	CONFIRMED	`Design.md` §6, §6b, §7, §10	Stale against the code: AddWire vs AddCurve; promised `report.html` not written; §10 outdated (V1–V6, no field.js or sim/, double-click claim); 400 vs 300 points; §7 “UNVALIDATED” label not implemented
D4	Medium	CONFIRMED	README, DEVELOPER_GUIDE, REQUEST_LOG, DESIGN_HISTORY, PROPOSAL_v2, AUDIT_BRAINSTORM_REVIEW, DECISIONS	Stale or missing: README says v1.5, has an old validation table and no v1.7 section; REQUEST_LOG has no v1.7 or audit entries; DESIGN_HISTORY says V15–V19; PROPOSAL says “nothing built”; the “Audit actions” section is missing; the second V15 run and a_eff are undocumented
D5	Medium	CONFIRMED	docs	`.html` missing or older than its `.md` for several documents; PDFs missing
D6	Medium	CONFIRMED	`dist/` , builder	`dist/` is v1.5; the builder omits newer docs; `dist/` is gitignored and unversioned
D7	Medium	CONFIRMED	NOTICE, README, SOURCES	Licence inventory lists Three.js only (missing Magpylib BSD-2, openEMS GPL-3 / CSXCAD LGPL-3, mariepy GPL,

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				numpy/scipy/h5py/torch); the NIST archive is missing from SOURCES.md
D8	Medium	CONFIRMED	`sim/v20_mariepy.py` , `tests/run.js`	V20 script untracked; run.js copies its verdict instead of applying the 10% criterion itself
D9	Medium	CONFIRMED	app, `tests/export_case.js` , PARAMETERS	Tissue and phantom constants duplicated; zMin untagged and absent from PARAMETERS; the app export always attaches a phantom
D10	Low	CONFIRMED	app	APP_VERSION says v1.6; the co-sim card omits warnings, git hash, V-verdicts; no check that results match the current design
D11	Low	LIKELY	app load and exports	Schema validated too late; missing fields become `undefined` ; partial state on error; no try/catch in exports
D12	Low	CONFIRMED	paths, `.gitignore`	Hard-coded user paths (overridable); npz policy undocumented; pre-M10 cosim outputs have no git hash
D13	Low	CONFIRMED	app	Accessibility: missing tab roles and labels; colour-only status; mouse-only 3D view
D14	Low	CONFIRMED	PARAMETERS	Contradictory zCut rows; self-L points 300 vs ≤ 800
D15	Low	CONFIRMED	Design §7, DECISIONS	Post-registration harness clarifications (M6, M13, M14) not logged in the project's own records
D16	Info	POSSIBLE	`docs/research_integrity/VALIDATION_PROTOCOL.md`	Cites hashes from another repository (Lab_3d)

Verified correct

- **Provenance headers** on all tracked source files.

- **Commit hashes:** all 14 cited hashes resolve, and the pre-registration commits contain what they claim.
- **Schemas:** the app's case export matches `tests/export_case.js` and every field the sim scripts read; tissue numbers are identical everywhere, with 3T labelled as an approximation.
- **Index mapping:** flat → wrapped mapping is sound (`S.wrap` keeps the point count).
- **Co-sim loader:** its keys match `cosim.py` .
- **Banner:** honest; any non-PASS is listed, and missing results read NOT RUN.
- **Workers:** the element worker guards stale results; the B1 worker restarts.
- **Labels:** exports carry exploratory labels.
- **Parameter tags:** γ/2π is LIT with an archived source; placeholders are tagged; the calibration file is tracked.